VIDEO ENCRYPTION AND DECRYPTION

Using the 5D tangent-hyperbolic memristive chaotic model

SUMMARY

The first 2.0 seconds of the supplied video were processed at 160 × 90 pixels and 30/1 fps. All 60 video frames and the extracted PCM audio were recovered exactly from the lossless encrypted MKV using the same chaotic key. This confirms a successful round trip for the processed preview, not for the complete original video.

1. Objective and research basis

The objective is to demonstrate video encryption and decryption using byte sequences generated from the five-dimensional tangent-hyperbolic memristive chaotic model. The video is separated into RGB frames and an audio track. Each part is encrypted independently with XOR and is decrypted by regenerating the same chaotic keystream.

The 2024 paper by Qureshi et al. introduces the 5D model and demonstrates an amalgamated image-encryption method. The frame-and-audio pipeline in this report is a project extension; it is not an audio/video algorithm claimed by that paper.

2. 5D chaotic model

The first equation places γ outside the hyperbolic tangent, as in the paper:

$$\dot{x}_1 = a(x_2 - x_1) + 0.25x_3 + g(\delta + \gamma \tanh(x_5))x_3$$

State	Differential equation
$\dot{x}_2$	$b\,x_1 - x_1x_3 + c\,x_2 - x_4$
$\dot{x}_3$	$d\,x_1x_2 - e\,x_3$
$\dot{x}_4$	x_1
$\dot{x}_5$	x_3

Parameter	Value	Initial condition	Value
a	78	$x_1(0)$	0
b	−16	$x_2(0)$	0
c	58	$x_3(0)$	8
d	35	$x_4(0)$	6
e	15	$x_5(0)$	0
g, δ, γ	1, 1, 0.5		

Numerical method: classical fourth-order Runge–Kutta (RK4), h = 0.001, with 10,000 transient steps discarded and 10 RK4 steps between output samples. This is the integer-order case α = 1. The notebook does not implement the paper's Atangana–Baleanu–Caputo fractional solver for α = 0.99.

3. Test video and processing scope

Property	Original uploaded video	Processed experiment
Input file	test data.mp4	First 2.0 seconds only

Property	Original uploaded video	Processed experiment
Dimensions	1920 × 1080	160 × 90 (resized preview)
Frame rate	30 fps	30 fps
Duration	12.843 seconds	2.0 seconds
Video codec	H.264	RGB frames encrypted; FFV1 used for lossless container
Audio	AAC, 48 kHz, 2 channels	PCM 16-bit, 48 kHz, 2 channels
Frame count	About 385 frames in full file	60 frames processed

The reduced duration and resolution were selected to complete a reproducible run in the available execution environment. Therefore, the measured results apply only to this two-second, 160 × 90 preview segment. They must not be described as a full-resolution or full-duration encryption result.

4. Encryption and decryption flow

- Read the MP4 and inspect its video and audio streams.
- Decode the first two seconds into RGB frames; resize each frame to 160 × 90.
- Extract the audio and convert it to 16-bit PCM while keeping 48 kHz and two channels.
- Initialize the 5D model using the fixed parameters and initial conditions; run RK4 and discard the transient.
- Generate a continuous chaotic stream for video frames. Convert x_1, x_2 and x_3 into three 8-bit masks for each pixel.
- XOR each RGB frame with its mask. The audio path starts a fresh chaotic stream and uses x_1 and x_2 to create 16-bit masks for the left and right channels.
- Save the encrypted frames and audio in an MKV container using FFV1 video and PCM audio, both lossless formats.
- Read the encrypted MKV, regenerate the same streams from the same initial conditions, and XOR again to decrypt.
- Compare every recovered frame and every recovered audio sample with the processed original arrays.

5. Encryption equations

For video pixels, the chaotic values are converted to 8-bit masks:

$$\mathbf{K_v = floor(|x| \times 10^8) \bmod 256}$$

$$\mathbf{C_v = P_v \text{ XOR } K_v \quad ; \quad \hat{P}_v = C_v \text{ XOR } K_v}$$

For 16-bit PCM audio, the values are converted to 16-bit masks:

$$\mathbf{K_a[n,c] = floor(|x^c[n]| \times 10^8) \bmod 65{,}536}$$

$$\mathbf{C_a[n,c] = P_a[n,c] \text{ XOR } K_a[n,c]}$$

XOR is reversible because (P XOR K) XOR K = P. The same initial state, model parameters, sampling order and stream-reset behavior must be used during decryption.

6. Actual output frames

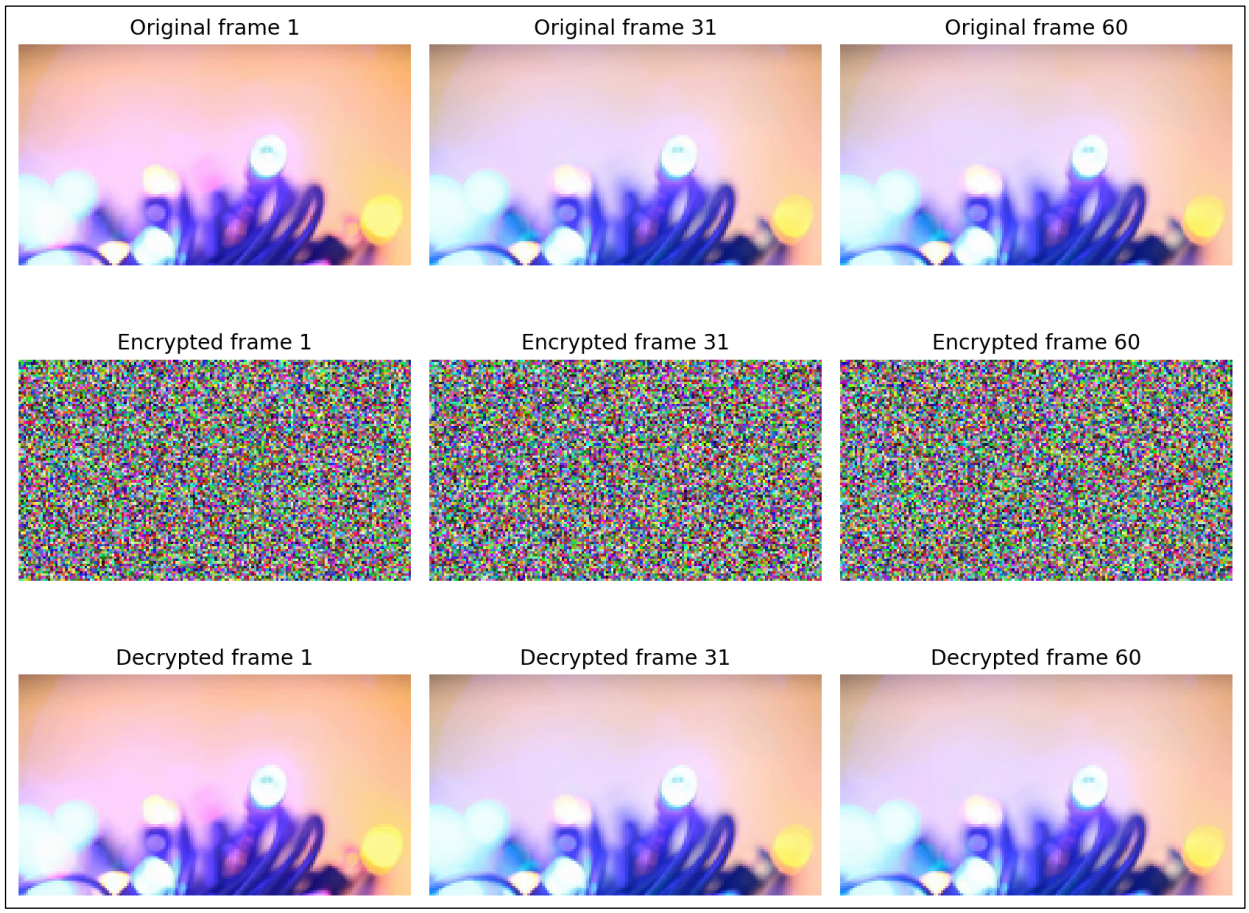


Figure 1. Original, encrypted and decrypted frames sampled from the beginning, middle and end of the processed two-second segment.

The encrypted frames appear noise-like. The recovered frames match the processed original frames exactly when compared as RGB arrays. The figure is retained in color; report text and tables use black and white.

7. Actual test results

Test	Measured result	Interpretation
Processed video frames	60	60 frames; all were recovered exactly
Frame recovery check	True	PASS — exact equality after lossless MKV read-back
Frame 1 MSE / PSNR	0.0 / ∞ dB	PASS — zero difference
Audio recovery check	True	PASS — all PCM samples recovered exactly
Audio MSE / SNR	0.0 / ∞ dB	PASS — zero difference
Original audio byte entropy	7.1693 bits	Measured over PCM bytes
Encrypted audio byte entropy	7.9996 bits	Close to the 8-bit maximum; statistical observation only
Temporal frame correlation	0.9307 → -0.0008	Mean over the first 20 consecutive-frame pairs; reduced strongly

Test	Measured result	Interpretation
Wrong-key audio SNR	-12.23 dB	Wrong initial condition does not recover the audio
Wrong-key frame 1 PSNR	6.97 dB	Wrong-key frame differs strongly from the original

8. Entropy, spatial correlation and wrong-key outputs

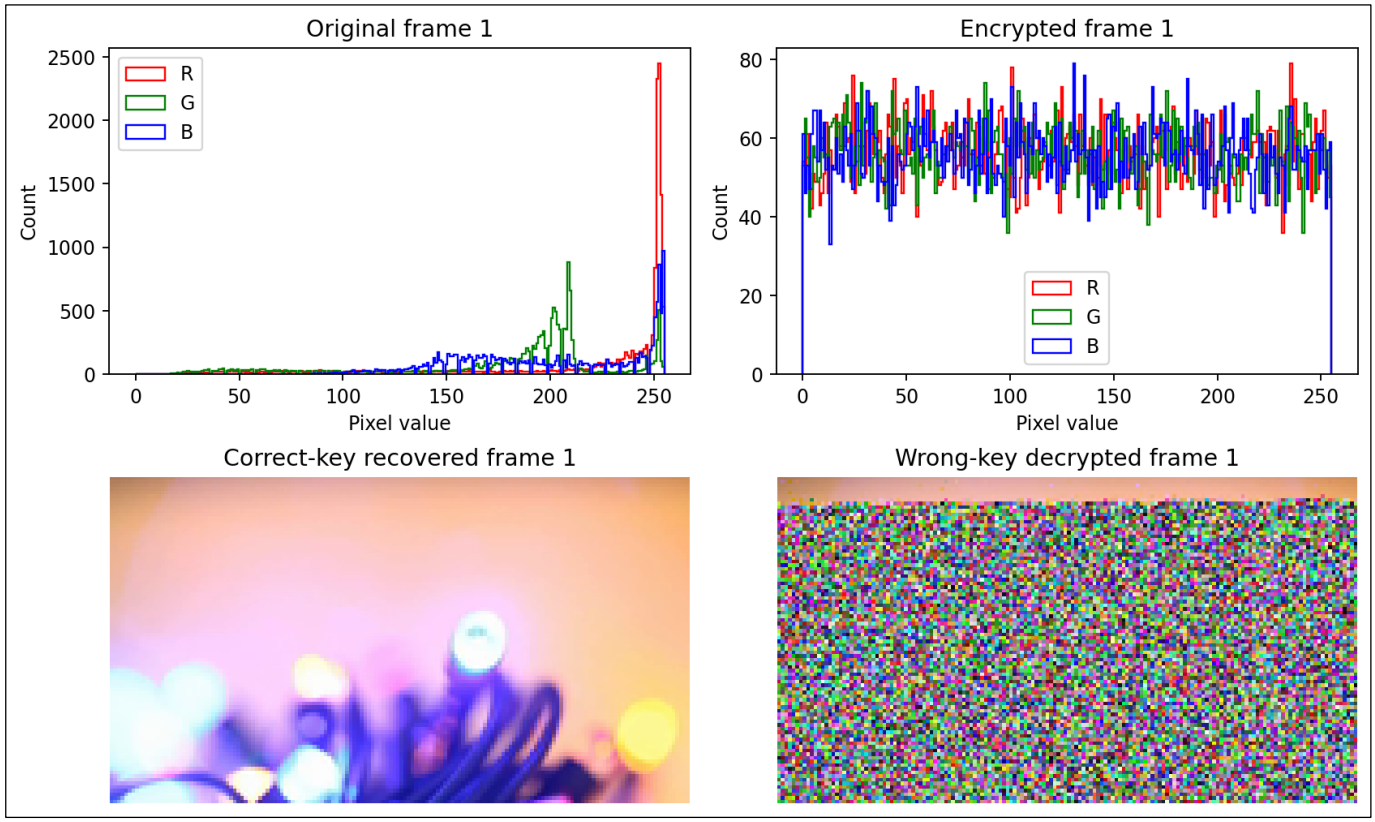


Figure 2. Original and encrypted frame-1 channel histograms, correct-key recovered frame and wrong-key decrypted frame.

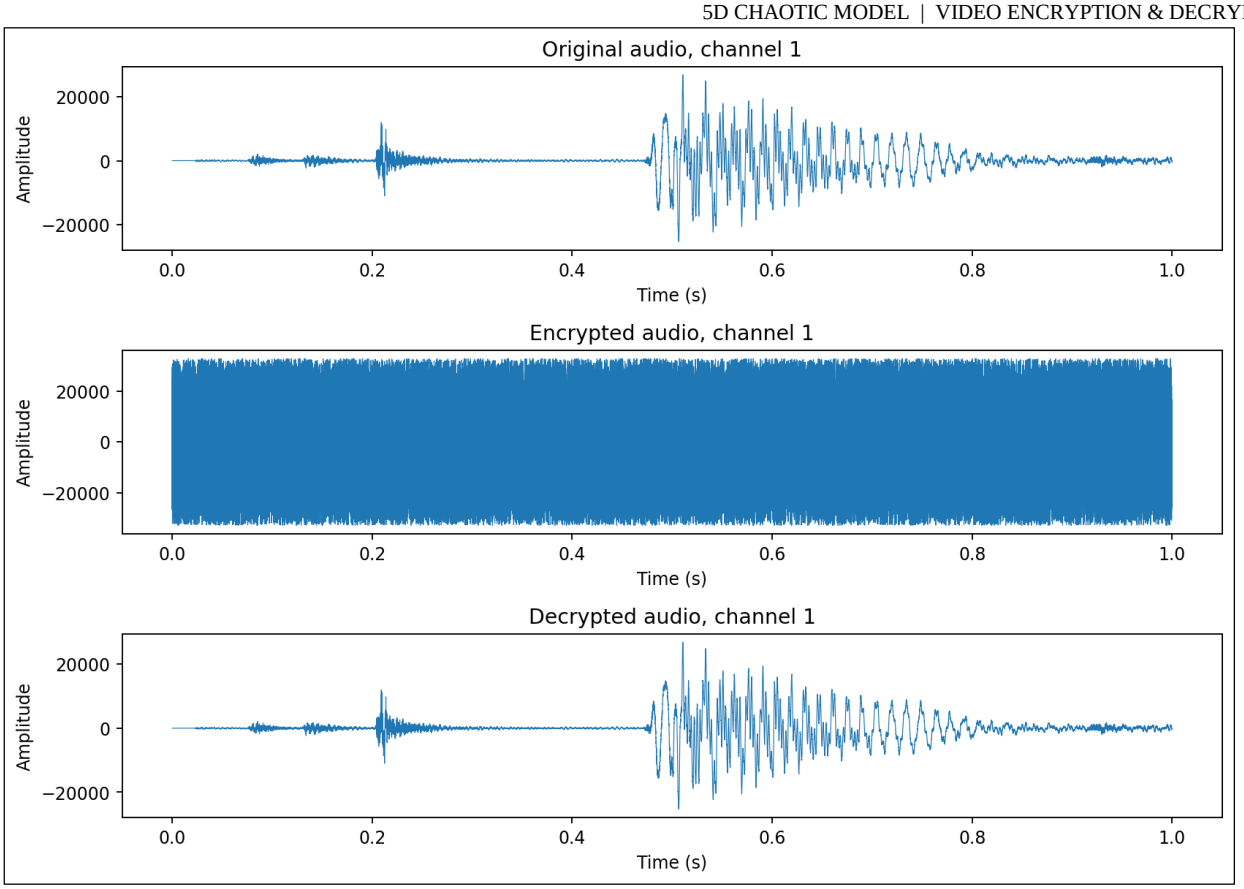


Figure 3. Original, encrypted and decrypted audio waveforms for the first two seconds (left channel).

For sampled frames 1, 31 and 60, each encrypted RGB-channel entropy is approximately 7.985–7.989 bits out of 8. Horizontal red-channel neighbour correlation changes from approximately 0.975–0.978 in the plain frames to values near zero in the encrypted frames. These observations describe the tested segment only and do not prove cryptographic security.

9. Important limitations and interpretation

- The run processes only the first two seconds and scales 1920 × 1080 frames down to 160 × 90. Full original duration and resolution were not processed.
- No separate key video, key image or key audio was provided. This run uses the chaotic keystream alone; optional second media keys are not used.
- The paper demonstrates the 5D chaotic model and image AIE method. This video-and-audio XOR pipeline is a project extension.
- RK4 implements $\alpha = 1$ only. The paper's ABC fractional-order $\alpha = 0.99$ scheme is not implemented in this run.
- The MKV output uses FFV1 and PCM to preserve encrypted data for decryption and exact comparison. The H.264/AAC MP4 files are lossy viewing previews and must not be used as the decryption source.
- The design is XOR-only and has no diffusion stage. A one-pixel change affects only the corresponding ciphertext pixel, so high NPCR/UACI values are not expected.
- The chaotic keystream is deterministic and restarts from the same fixed key. Reusing it across different files can expose relationships between plaintexts. This prototype is not proven secure and should not be used for confidential production data.

10. Conclusion

The 5D model was used to encrypt the first 2.0 seconds of the supplied video at 160 × 90 and 30/1 fps. The lossless MKV round-trip recovered all 60 processed RGB frames and all 96,000 stereo PCM sample frames exactly (frame MSE = 0; audio MSE = 0). Entropy and correlation changed in the expected statistical direction, and wrong-key tests did not recover the original. These results verify reversible processing of the reduced preview segment; they do not establish production-grade security or full-video/full-resolution performance.

Reference

Qureshi, M. A., Khan, N. A., Raza, S., & Iqbal, S. M. Z. (2024). “Cap like trajectories in 5D chaotic tangent hyperbolic memristive model: fractional calculus and encryption.” Physica Scripta, 99, 075238. https://doi.org/10.1088/1402-4896/ad5476

Github link:

https://github.com/fizza-14/5D-Chaotic-Memristive-Image-Audio-Video-Encryption-Decryption